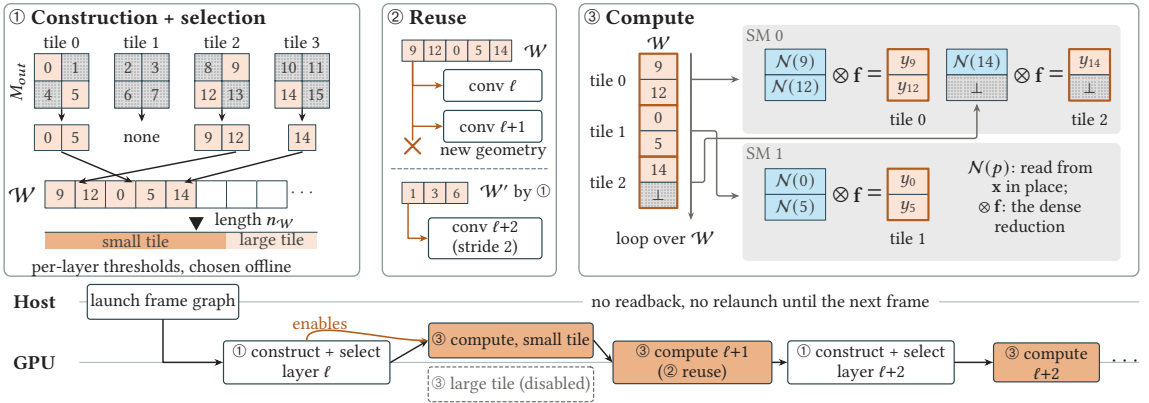

Ours: WISEConv
① Construction + selection
tile 0
tile 1
tile 2
tile 3
M_{out}
0 1 4 5
2 3 6 7
8 9 12 13
10 11 14 15
0 5
none
9 12
14
$\mathcal{W}$ 9 12 0 5 14 · · ·
length $n_{\mathcal{W}}$
small tile
large tile
per-layer thresholds, chosen offline
② Reuse
9 12 0 5 14 $\mathcal{W}$
conv ℓ
conv ℓ+1
new geometry
1 3 6 $\mathcal{W}'$ by ①
conv ℓ+2
(stride 2)
③ Compute
$\mathcal{W}$
tile 0
tile 1
tile 2
9 12 0 5 14 ⊥
loop over $\mathcal{W}$
SM 0
$\mathcal{N}(9)$ $\mathcal{N}(12)$ ⊗f = y_9 y_{12}
tile 0
$\mathcal{N}(14)$ ⊥ ⊗f = y_{14} ⊥
tile 2
SM 1
$\mathcal{N}(0)$ $\mathcal{N}(5)$ ⊗f = y_0 y_5
tile 1
$\mathcal{N}(p)$: read from x in place;
⊗f: the dense reduction
Host
launch frame graph
no readback, no relaunch until the next frame
GPU
enables
① construct + select layer ℓ
③ compute, small tile
③ large tile (disabled)
③ compute ℓ+1 (② reuse)
① construct + select layer ℓ+2
③ compute ℓ+2
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